

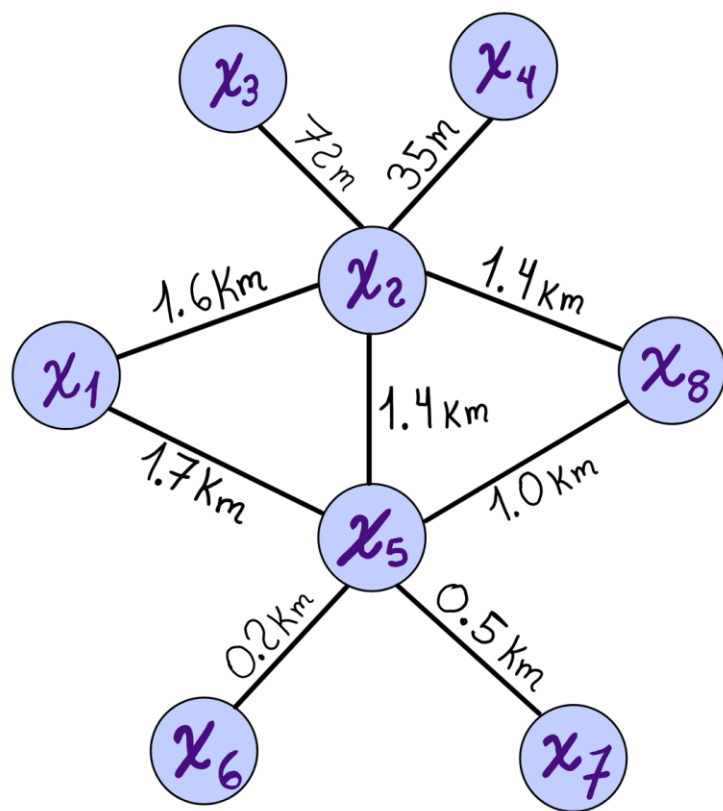
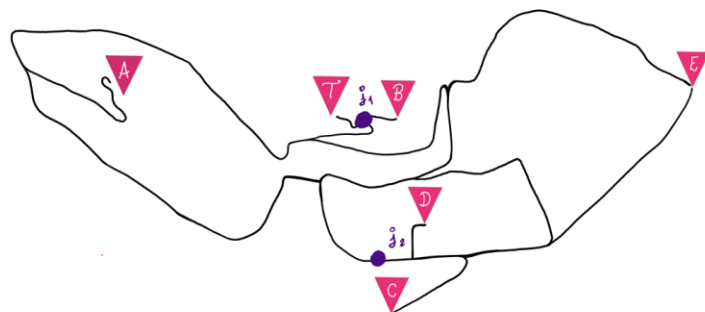


Ronaldo & Rodrigo

QuantumQuipu Hackathon

# OPTIMIZACIÓN DE RUTAS EN MINERÍA

Una aproximación a la minería desde algoritmos cuánticos variacionales



$x_1$  : Planta de procesamiento

$x_2$  :  $j_1$

$x_3$  : Tajo

$x_4$  : Botadero

$x_5$  :  $j_2$

$x_6$  : Lixiviación

$x_7$  : Baja ley

$x_8$  : Relaves

$d_{12}$  (1,6 km)

$d_{23}$  (72 m)

$d_{24}$  (35 m)

$d_{28}$  (1,4 km)

$d_{25}$  (1,4 km)

$d_{15}$  (1,7 km)

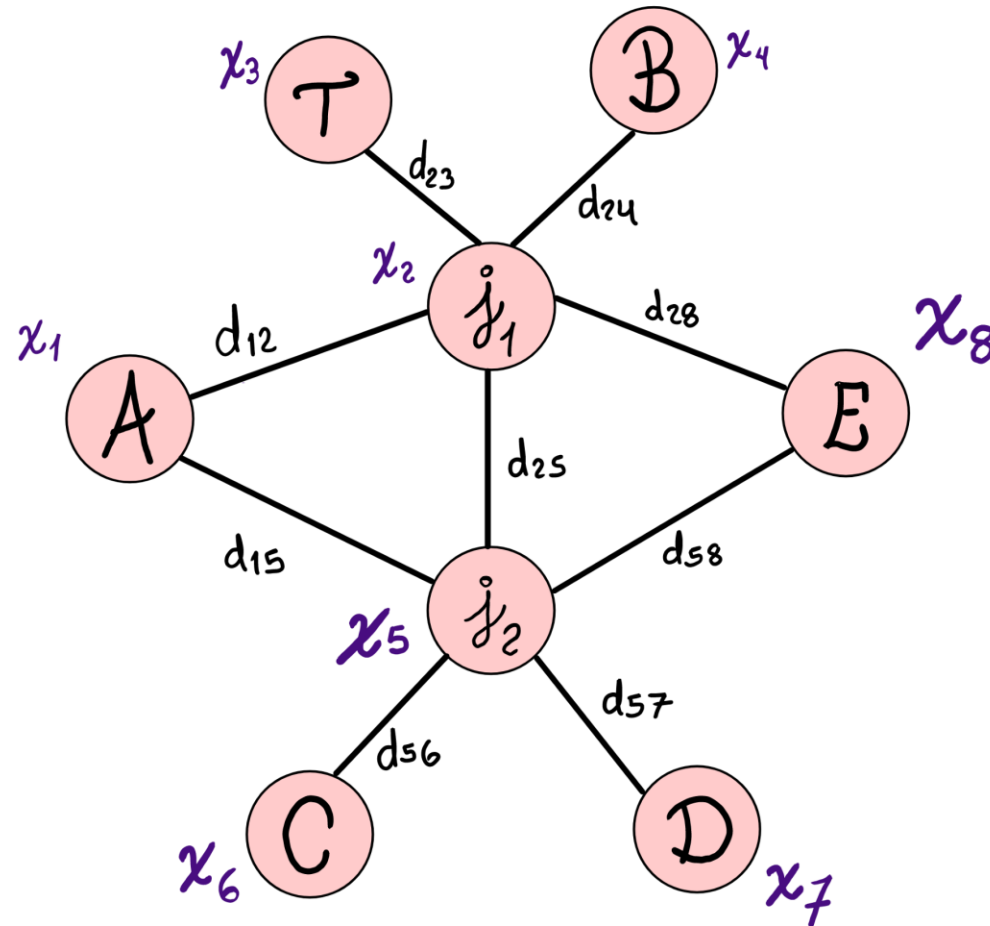
$d_{85}$  (1,0 km)

$d_{56}$  (0,2 km)

$d_{57}$  (0,5 km)

# OPTIMIZACIÓN POR QUBO

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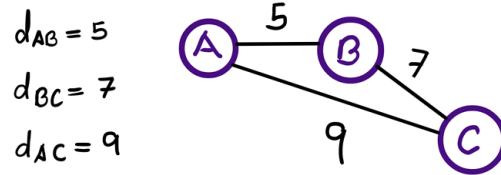
## Ejemplo rápido: Trayectoria mínima

función de coste:

$$Q_C = B \sum_{i,j}^N d_{ij} \sum_t^T x_{i,t} x_{j,t+1}$$

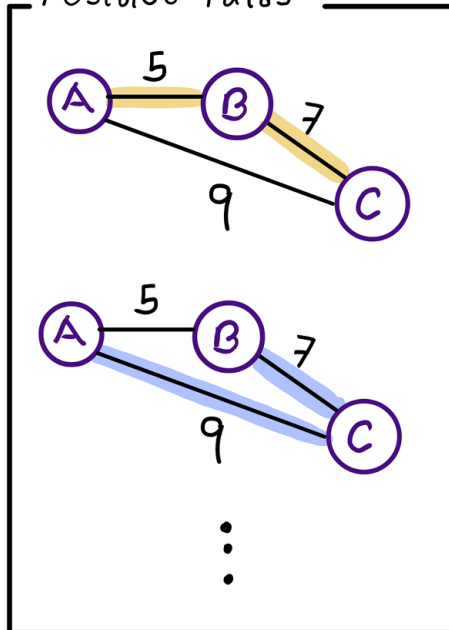
$x_{i,t} \begin{cases} 1, & \text{Si el camión visita} \\ & \text{la ubicación } i \text{ en el paso } t. \\ 0 \end{cases}$

Usando rutas pequeñas:



\* Naturalmente  
la solución  
evita los  
castigos.

Posibles rutas



$$Q_C^1 = B \sum_{j,i}^3 d_{ij} \sum_t^2 x_{i,t} x_{j,t+1} = d_{12} + d_{23} = 5 + 7 = 12$$

$$\begin{aligned} & \circ d_{11}(x_{1,1}x_{1,2} + x_{1,2}x_{1,3}) + d_{21}(x_{2,1}x_{1,2} + x_{2,2}x_{1,3}) + d_{31}(x_{3,1}x_{1,2} + x_{3,2}x_{1,3}) \\ & + d_{12}(x_{1,1}x_{2,2} + x_{1,2}x_{2,3}) + d_{22}(x_{2,1}x_{2,2} + x_{2,2}x_{2,3}) + d_{32}(x_{3,1}x_{2,2} + x_{3,2}x_{2,3}) \\ & + d_{13}(x_{1,1}x_{3,2} + x_{1,2}x_{3,3}) + d_{23}(x_{2,1}x_{3,2} + x_{2,2}x_{3,3}) + d_{33}(x_{3,1}x_{3,2} + x_{3,2}x_{3,3}) \end{aligned}$$

$$Q_C^2 = B \sum_{j,i}^3 d_{ij} \sum_t^2 x_{i,t} x_{j,t+1} = d_{13} + d_{32} = 9 + 7 = 16$$

$$\begin{aligned} & \circ d_{11}(x_{1,1}x_{1,2} + x_{1,2}x_{1,3}) + d_{21}(x_{2,1}x_{1,2} + x_{2,2}x_{1,3}) + d_{31}(x_{3,1}x_{1,2} + x_{3,2}x_{1,3}) \\ & + d_{12}(x_{1,1}x_{2,2} + x_{1,2}x_{2,3}) + d_{22}(x_{2,1}x_{2,2} + x_{2,2}x_{2,3}) + d_{32}(x_{3,1}x_{2,2} + x_{3,2}x_{2,3}) \\ & + d_{13}(x_{1,1}x_{3,2} + x_{1,2}x_{3,3}) + d_{23}(x_{2,1}x_{3,2} + x_{2,2}x_{3,3}) + d_{33}(x_{3,1}x_{3,2} + x_{3,2}x_{3,3}) \end{aligned}$$

sol:

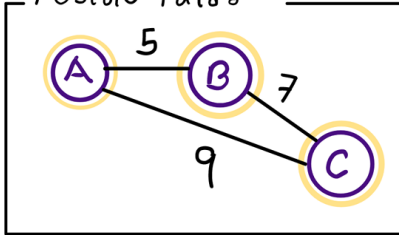
|          | $t_1$ | $t_2$ | $t_3$ |
|----------|-------|-------|-------|
| $A(x_1)$ | 1     | 0     | 0     |
| $B(x_2)$ | 0     | 1     | 0     |
| $C(x_3)$ | 0     | 0     | 1     |

|          | $t_1$ | $t_2$ | $t_3$ |
|----------|-------|-------|-------|
| $A(x_1)$ | 1     | 0     | 0     |
| $B(x_2)$ | 0     | 0     | 1     |
| $C(x_3)$ | 0     | 1     | 0     |

sol<sup>o</sup>

|          | $t_1$ | $t_2$ | $t_3$ |
|----------|-------|-------|-------|
| $A(x_1)$ | 1     | 0     | 0     |
| $B(x_2)$ | 1     | 0     | 0     |
| $C(x_3)$ | 1     | 0     | 0     |

Posible ruta:



$$Q_C^1: B \sum_{j,i}^3 dij \sum_t^2 x_{i,t} x_{j,t+1} = 0$$

$$\begin{aligned} & d_{11}(\cancel{x_{1,1}x_{1,2}} + \cancel{x_{1,2}x_{1,3}}) + d_{21}(\cancel{x_{2,1}x_{1,2}} + \cancel{x_{2,2}x_{1,3}}) + d_{31}(\cancel{x_{3,1}x_{1,2}} + \cancel{x_{3,2}x_{1,3}}) \\ & + d_{12}(\cancel{x_{1,1}x_{2,2}} + \cancel{x_{1,2}x_{2,3}}) + d_{22}(\cancel{x_{2,1}x_{2,2}} + \cancel{x_{2,2}x_{2,3}}) + d_{32}(\cancel{x_{3,1}x_{2,2}} + \cancel{x_{3,2}x_{2,3}}) \\ & + d_{13}(\cancel{x_{1,1}x_{3,2}} + \cancel{x_{1,2}x_{3,3}}) + d_{23}(\cancel{x_{2,1}x_{3,2}} + \cancel{x_{2,2}x_{3,3}}) + d_{33}(\cancel{x_{3,1}x_{3,2}} + \cancel{x_{3,2}x_{3,3}}) \end{aligned}$$

Restricción Temporal:

$$Q_B = A \sum_{t=1}^T \left( 1 - \sum_{i=1}^N x_{i,t} \right)^2$$

Para que combinación no se tome en cuenta para la minimización  $A \gg B$

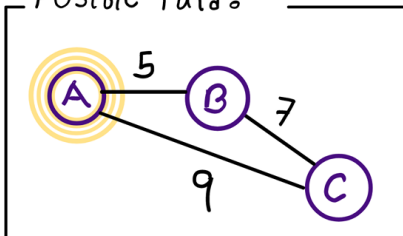
$$\begin{aligned} & \sum_t (1 - x_{1,t} - x_{2,t} - x_{3,t})^2 \\ & (1 - \cancel{x_{1,1}} - \cancel{x_{2,1}} - \cancel{x_{3,1}})^2 \rightarrow (1-3)^2 = 4 \\ & + (1 - \cancel{x_{1,2}} - \cancel{x_{2,2}} - \cancel{x_{3,2}})^2 \rightarrow 1^2 \\ & + (1 - \cancel{x_{1,3}} - \cancel{x_{2,3}} - \cancel{x_{3,3}})^2 \rightarrow 1^2 \\ & Q_B = 6A \end{aligned}$$

$x_{it}$ , diferentes lugares en un mismo tiempo.

sol<sup>o</sup>

|          | $t_1$ | $t_2$ | $t_3$ |
|----------|-------|-------|-------|
| $A(x_1)$ | 1     | 1     | 1     |
| $B(x_2)$ | 0     | 0     | 0     |
| $C(x_3)$ | 0     | 0     | 0     |

Posible ruta:



$$Q_C^2: B \sum_{j,i}^3 dij \sum_t^2 x_{i,t} x_{j,t+1} = d_{11} + d_{11} = 0$$

$$\begin{aligned} & d_{11}(\cancel{x_{1,1}x_{1,2}} + \cancel{x_{1,2}x_{1,3}}) + d_{21}(\cancel{x_{2,1}x_{1,2}} + \cancel{x_{2,2}x_{1,3}}) + d_{31}(\cancel{x_{3,1}x_{1,2}} + \cancel{x_{3,2}x_{1,3}}) \\ & + d_{12}(\cancel{x_{1,1}x_{2,2}} + \cancel{x_{1,2}x_{2,3}}) + d_{22}(\cancel{x_{2,1}x_{2,2}} + \cancel{x_{2,2}x_{2,3}}) + d_{32}(\cancel{x_{3,1}x_{2,2}} + \cancel{x_{3,2}x_{2,3}}) \\ & + d_{13}(\cancel{x_{1,1}x_{3,2}} + \cancel{x_{1,2}x_{3,3}}) + d_{23}(\cancel{x_{2,1}x_{3,2}} + \cancel{x_{2,2}x_{3,3}}) + d_{33}(\cancel{x_{3,1}x_{3,2}} + \cancel{x_{3,2}x_{3,3}}) \end{aligned}$$

$$d_{11}, d_{22}, d_{33} = 0$$

Restricción Espacial:

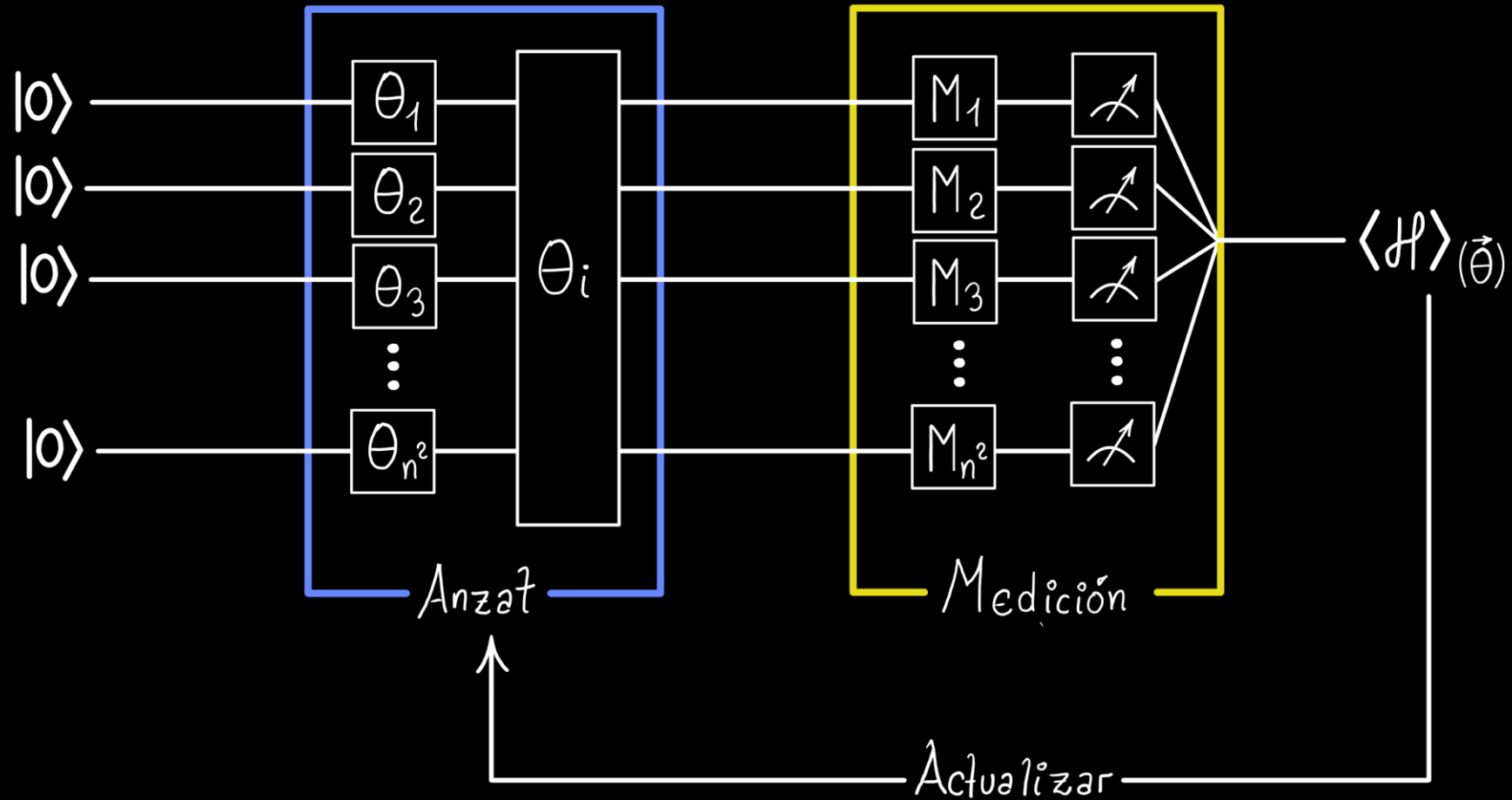
$$Q_A = A \sum_{i=1}^N \left( 1 - \sum_{t=1}^T x_{i,t} \right)^2$$

$$\begin{aligned} & \sum_{i=1}^N (1 - x_{i1} - x_{i2} - x_{i3})^2 \\ & (1 - \cancel{x_{11}} - \cancel{x_{12}} - \cancel{x_{13}})^2 \rightarrow (1-3)^2 = 4 \\ & (1 - \cancel{x_{21}} - \cancel{x_{22}} - \cancel{x_{23}})^2 \rightarrow 1^2 \\ & (1 - \cancel{x_{31}} - \cancel{x_{32}} - \cancel{x_{33}})^2 \rightarrow 1^2 \\ & Q_A = 6A \end{aligned}$$

$x_{it}$ , mismo lugar en tiempos diferentes.

# OPTIMIZACIÓN POR VQE

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## Ejemplo rápido: Ciclo de optimización VQE

Escalar  $\rightarrow$   $\mathcal{Q}$  función de Coste:

$$\mathcal{Q}(x) = B \sum_{i,j} d_{ij} \sum_{\tau} x_{i,\tau} x_{j,\tau+1} + \underbrace{A \sum_{\tau=1}^T \left(1 - \sum_{i=1}^N x_{i,\tau}\right)^2}_{Q_{\text{temporal}}} + \underbrace{A \sum_{i=1}^N \left(1 - \sum_{\tau=1}^T x_{i,\tau}\right)^2}_{Q_{\text{espacial}}}$$

$N = 3$  nodos  
 $T = 3$  tiempos

$x_{i,\tau} \in \{0,1\}$

Variables binarias  $x_{i,\tau} \in \{x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}, x_{31}, x_{32}, x_{33}\}$

**Mapeo Ising**  
 (Vamos a recomodar el  $\mathcal{Q}$  para que pueda ser leído por VQE)

Un Qubit no sabe que es 0 y 1.

$$\begin{aligned} Z|0\rangle &= +1|0\rangle \\ Z|1\rangle &= -1|1\rangle \end{aligned}$$

**Mapeo estándar**  
 $x=0 \rightarrow z=1$   
 $x=1 \rightarrow z=-1$

$$\boxed{x_{i,\tau} = \frac{1 - z_{i,\tau}}{2}} \quad | \quad z_{i,\tau} \in \{-1, +1\}$$

$\mathcal{Q}(z)$   $\rightarrow$   $\mathcal{H}$  función de Coste (Ising):

$$\mathcal{H}(z) = B \left\{ \frac{T-1}{4} \sum_{i,j} d_{ij} - \frac{1}{4} \sum_{i=1}^N \left[ a_i z_{i,1} + b_i z_{i,T} + \sum_{\tau=2}^{T-1} (a_i + b_i) z_{i,\tau} \right] + \frac{1}{4} \sum_{\tau=1}^{T-1} \sum_{i,j} d_{ij} z_{i,\tau} z_{j,\tau+1} \right\} +$$

$\uparrow$   
función no cuántica

$$A \left\{ \frac{T}{4} (4 - 3N + N^2) + (1 - N/2) \sum_i \sum_{\tau} z_{i,\tau} + \frac{1}{2} \sum_{\tau} \sum_{i,j} s_{i,\tau} s_{j,\tau} \right\} + A \left\{ \frac{N}{4} (4 - 3T + T^2) + (1 - T/2) \sum_i \sum_{\tau} z_{i,\tau} + \frac{1}{2} \sum_{\tau} \sum_{i < j} s_{i,\tau} s_{j,\tau} \right\}$$

$Q_{\text{temporal}}$ 
 $Q_{\text{espacial}}$

$$x_{i,t} \cdot x_{j,t+1} = \frac{1}{4} (1 - z_{i,t} - z_{j,t+1} + z_{i,t} z_{j,t+1}) \quad (1 - \sum_i x_{i,t})^2 = 1 - 2 \sum_i \underbrace{x_{i,t}} + \sum_{i,j} \underbrace{x_{i,t} x_{j,t}}$$

$$x_{i,t} \cdot x_{i,t'} = \frac{1}{4} (1 - z_{i,t} - z_{i,t'} + z_{i,t} z_{i,t'}) \quad (1 - \sum_i x_{i,t})^2 = 1 - 2 \sum_t \underbrace{x_{i,t}} + \sum_{t,t'} \underbrace{x_{i,t} x_{i,t'}}$$

$$Q(x) = \underbrace{\frac{B}{4} \sum_{i,j} d_{ij} \sum_t^{T-1} (1 - z_{i,t} - z_{j,t+1} + z_{i,t} z_{j,t+1})}_{Q_{\text{temporal}}} + \underbrace{A \sum_{t=1}^T \left(1 - \sum_{i=1}^N x_{i,t}\right)^2 + A \sum_{i=1}^N \left(1 - \sum_{t=1}^T x_{i,t}\right)^2}_{Q_{\text{especial}}}$$

$$\frac{1}{4} \sum_{i,j} d_{ij} \sum_t^{T-1} (1 - z_{i,t} - z_{j,t+1} + z_{i,t} z_{j,t+1})$$

$$\frac{T-1}{4} \sum_{i,j} d_{ij} + \frac{1}{4} \sum_{i,j} \sum_t^{T-1} (-d_{ij} z_{i,t} - d_{ij} z_{j,t+1} + d_{ij} z_{i,t} z_{j,t+1})$$

$$a_i = \sum_j d_{ij}$$

$$b_j = \sum_i d_{ij}$$

$$\frac{1}{4} \sum_i \sum_t^{T-1} a_i z_{i,t} + \frac{1}{4} \sum_j \sum_t^{T-1} b_j z_{j,t+1}$$

$$\text{con } t=2$$

$$+ \frac{1}{4} \sum_j \sum_{t=2}^T b_j z_{j,t}$$

$$\frac{1}{4} \sum_{t=2}^T \sum_i a_i z_{i,t} + \frac{1}{4} \sum_i a_i z_{i,1} + \frac{1}{4} \sum_j \sum_{t=2}^{T-1} b_j z_{j,t} + \frac{1}{4} \sum_j b_j z_{j,T}$$

usamos  $j=i$   
(dummy)

$Q_c$ :

$$B \left\{ \underbrace{\frac{T-1}{4} \sum_{i,j} d_{ij}}_{cst \in} - \frac{1}{4} \sum_{i=1}^N \left[ a_i z_{i,1} + b_i z_{i,T} + \sum_{t=2}^{T-1} (a_i + b_i) z_{i,t} \right] + \frac{1}{4} \sum_{t=1}^{T-1} \sum_{i,j} d_{ij} z_{i,t} z_{j,t+1} \right\}$$

$$h_{ij}^{(c)} = \frac{B}{4} \times \begin{cases} a_i z_{i,1}, & t=1 \\ a_i + b_i = \sum_j (p_{ij} + p_{ji}), & 2 \leq t \leq T-1 \\ b_i z_{i,T}, & t=T \end{cases}$$

acoplamiento

$$\gamma_{(i,t)(j,t+1)}^{(c)} = \frac{B}{4} d_{ij}$$

$$Q_{\text{esp}}: \sum_{i=1}^N \left(1 - \sum_{t=1}^T x_{i,t}\right)^2$$

$$\sum_{i=1}^N \left(1 - 2 \sum_t x_{i,t} + \sum_{t,t'} x_{i,t} x_{i,t'}\right)$$

$$\sum_{i=1}^N \left(1 - T + \sum_t z_{i,t}\right) + \sum_i \sum_{t,t'} \frac{1}{4} (1 - z_{i,t} - z_{i,t'} + z_{i,t} z_{i,t'})$$

$$\sum_{i=1}^N \left(1 - T + \sum_t z_{i,t}\right) + \sum_i \sum_{t,t'} \frac{1}{4} - \frac{T}{4} \sum_i \sum_t z_{i,t} - \frac{T}{4} \sum_i \sum_{t'} z_{i,t'} + \frac{1}{4} \sum_i \sum_{t,t'} z_{i,t} z_{i,t'}$$

$$N - TN + \frac{NT^2}{4} + \sum_i \sum_t z_{i,t} - \frac{T}{4} \sum_i \sum_t z_{i,t} - \frac{T}{4} \sum_i \sum_{t'} z_{i,t'}$$

$$N(1 - T + \frac{T^2}{4}) + \sum_i \sum_t z_{i,t} - \frac{T}{2} \sum_i \sum_t z_{i,t} + \frac{1}{4}$$

$$N(1 - T + \frac{T^2}{4}) + \left(1 - \frac{T}{2}\right) \sum_i \sum_t z_{i,t}$$

$$\rightarrow \left(-NT + \frac{1}{4} NT^2\right)$$

$$N \left(1 - \frac{3}{4} T + \frac{T^2}{4}\right)$$

$$+ \frac{1}{4} \sum_i \sum_{t,t'} z_{i,t} z_{i,t'} \quad z_{i,t}^2 = 1$$

$$\frac{1}{4} \sum_i \left( \sum_{t,t'} z_{i,t} z_{i,t'} + \sum_{t,t'} s_{i,t} s_{i,t'} \right)$$

$$\frac{1}{4} \sum_i \sum_t 1 + \frac{1}{4} \sum_i \sum_{t,t'} s_{i,t} s_{i,t'} \quad \text{aparece } \sum_{t,t'} s_{i,t} s_{i,t'} = 2 \sum_{t < t'} s_{i,t} s_{i,t'}$$

$$\frac{1}{4} TN + \frac{1}{4} 2 \sum_i \sum_{t < t'} s_{i,t} s_{i,t'}$$

$$Q_{\text{esp}}: A \left[ \underbrace{\frac{N(4 - 3T + T^2)}{4}}_{cst \in} + \left(1 - \frac{T}{2}\right) \sum_i \sum_t z_{i,t} + \frac{1}{2} \sum_i \sum_{t < t'} s_{i,t} s_{i,t'} \right]$$

$$h_{i,t}^{(esp)} = A(1 - T/2) \quad \gamma_{(i,t)(j,t)} = \frac{A}{2} \quad (i < j \text{ mismo } t)$$

$$Q_{\text{tem}}: A \left[ \underbrace{\frac{T(4 - 3N + N^2)}{4}}_{cst \in} + \left(1 - \frac{N}{2}\right) \sum_i \sum_t z_{i,t} + \frac{1}{2} \sum_t \sum_{i < j} s_{i,t} s_{j,t} \right]$$

$$h_{i,t}^{(tem)} = A(1 - N/2) \quad \gamma_{(i,t)(i,t')} = \frac{A}{2} \quad (t < t' \text{ mismo } i)$$

# Función de Coste (Ising) :

$$\mathcal{H}(z) = B \left\{ \frac{T-1}{4} \sum_{i,j}^N d_{ij} - \frac{1}{4} \sum_{i=1}^N \left[ a_i z_{i,1} + b_i z_{i,T} + \sum_{t=2}^{T-1} (a_i + b_i) z_{i,t} \right] + \frac{1}{4} \sum_{t=1}^{T-1} \sum_{i,j}^N d_{ij} z_{i,t} z_{j,t+1} \right\} + Q_{esp}$$

$$Q_t A \left\{ \frac{T}{4} (4 - 3N + N^2) + (1 - N/2) \sum_{i,j}^N z_{i,t} + \frac{1}{2} \sum_{i,j}^N s_{i,t} s_{j,t} \right\} + A \left\{ \frac{N}{4} (4 - 3T + T^2) + (1 - T/2) \sum_{i,j}^T z_{i,t} + \frac{1}{2} \sum_{i,j}^T s_{i,t} s_{j,t} \right\}$$

$$Cst \text{ (Que se ignora)} = B \left( \frac{T-1}{4} \sum_{i,j}^N d_{ij} \right) + A \frac{T(N^2 - 3N + 4)}{4} + A \frac{N(T^2 - 3T + 4)}{4}$$

## Campos locales

$$h_{it} = \underbrace{B h_{it}^{(c)}}_{\text{término de costo}} + \underbrace{A(1 - N/2)}_{\text{restricción temporal}} + \underbrace{A(1 - T/2)}_{\text{restricción espacial}}$$

$$h_{it}^{(c)} = -\frac{1}{4} \times \begin{cases} \sum_{j=1}^N d_{ij} & , t=1 \\ \sum_{j=1}^N d_{ij} + d_{ji} & , 2 \leq t \leq T-1 \\ \sum_{j=1}^N d_{ji} & , t=T \end{cases}$$

## Campo alineado

$$h_{ij} > 0 : z \uparrow \sim +1$$

favorece  $x=0$

$$h_{ij} < 0 : z \downarrow \sim -1$$

## Acoplamientos

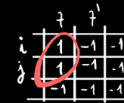
• transición  $Q_c$  :

$$J_{(i,t)(j,t+1)} = \frac{B}{4} d_{ij} \leftarrow \text{Costo de transición}$$

Para  $t=1, \dots, T-1, \forall i, j$

• mismo tiempo  $Q^{temp}$  :

$$J_{(i,t)(j,t)} = \frac{A}{2} \leftarrow \text{Penalización se activa con } i, j \text{ igual } t$$



• mismo espacio  $Q^{esp}$  :

$$J_{(i,t)(i,t')} = \frac{A}{2} \leftarrow \text{Penalización se activa con } t, t' \text{ igual } i$$



$$\mathcal{H} = Cste + h_{it} \sum_i^N \sum_t^T z_{i,t} + \sum_{t=1}^{T-1} \sum_{i,j}^N J^c d_{ij} z_{i,t} z_{j,t+1} + \sum_t^T \sum_{i,j}^N J^{temp} s_{i,t} s_{j,t} + \sum_i^N \sum_{t < t'}^T J^{esp} s_{i,t} s_{i,t'}$$

AA.

*H: Es una función matemática (matriz de coste) que mapea configuraciones de spins a valores de energía.*

Nuestra propuesta aún clásica.

## Mapeo Ising

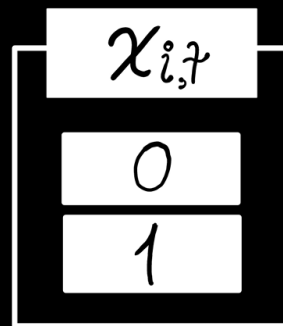
$$\mathcal{H} = \text{conste} + \sum_{i=1}^N \sum_{\tau=1}^T h_{i\tau} z(i, \tau) + \sum_{\text{pares únicos}} \mathcal{J}_{(i,\tau)(j,\tau')} z(i, \tau) z(j, \tau')$$

$$\begin{aligned} &+ A/2 \sum_{\tau=1}^T \sum_{1 \leq i < j \leq N} z(i, \tau) z(j, \tau) \\ &+ A/2 \sum_{i=1}^N \sum_{1 \leq \tau < \tau' \leq T} z(i, \tau) z(i, \tau') \\ &+ B/4 \sum_{\tau=1}^{T-1} \sum_{i=1}^N \sum_{j=1}^N d_{ij} \cdot z(i, \tau) z(j, \tau+1) \end{aligned}$$

Qubit

Bit

|          | $\tau_1$ | $\tau_2$ | $\tau_3$ |
|----------|----------|----------|----------|
| $A(x_1)$ | 1        | 0        | 0        |
| $B(x_2)$ | 0        | 1        | 0        |
| $C(x_3)$ | 0        | 0        | 1        |



$\Rightarrow$

$z_{i,\tau}$

-1

+1

$\nabla_z$

| $i \setminus \tau$ | $\tau_1$ | $\tau_2$ | $\tau_3$ |
|--------------------|----------|----------|----------|
| $i=1$              | ■        | •        | •        |
| $i=2$              | •        | ■        | •        |
| $i=3$              | •        | •        | ■        |

• := -1, ■ := +1

$$\mathcal{H} = \text{cste} + \sum_{i=1}^N \sum_{t=1}^T h_{it} z(i,t) + \sum_{\text{pares \u00fanicos}} \mathcal{T}_{(i,t)(j,t')} z(i,t) z(j,t')$$

$M = N \times T$  Variables

$\Rightarrow k = (i-1) \cdot T + t, K = 1, \dots, M$  (demostrar)

$$\mathcal{H}_{\text{matriz}} = \sum_{k=1}^M h_k z_k + \underbrace{\sum_{\ell=k+1}^M \sum_{k=1}^M}_{k < \ell} \mathcal{T}_{k\ell} z_k z_\ell$$

$\mathcal{T}_{(i,t)(j,t')}$



$\mathcal{T}_{k\ell}$

$$\mathcal{H} = \mathbf{z}^T \mathbf{J} \mathbf{z} + \mathbf{h}^T \mathbf{z}$$

$$\mathcal{H} = \mathbf{Z}^T \mathbf{J} \mathbf{Z} + \mathbf{h}^T \mathbf{Z}$$

2. Para nuestro ejemplo:

$$(N=3, T=3), K=(i-1) \cdot T + t$$

$$M=9$$

$$\mathbf{z}_{11} \rightarrow K=1$$

$$\mathbf{z}_{12} \rightarrow K=2$$

$$\mathbf{z}_{13} \rightarrow K=3$$

$$\mathbf{z}_{21} \rightarrow K=4$$

$$\mathbf{z}_{22} \rightarrow K=5$$

$$\mathbf{z}_{23} \rightarrow K=6 \quad [\dots (2-1) \cdot 3 + 3 = 6]$$

$$\mathbf{z}_{31} \rightarrow K=7$$

$$\mathbf{z}_{32} \rightarrow K=8$$

$$\mathbf{z}_{33} \rightarrow K=9 \quad [\dots (3-1) \cdot 3 + 3 = 9]$$

$$\mathcal{J}_{(i,t)(j,t+1)} = \frac{B}{4} d_{ij} \quad \begin{matrix} \nearrow \text{simetrice} \\ \Rightarrow \mathcal{J}_{sim} \end{matrix}$$

$$K=(i-1)T+t$$

$$l=(j-1)T+t+1$$

$$\mathcal{J}_{Kl} = (\dots)_{9 \times 9}$$

$$h_K \Rightarrow h_{it} \Rightarrow$$

$$B \underbrace{h_{it}^{(c)}} + A(1 - N/2) + A(1 - T/2)$$

$$\gamma_{it}^{(c)} = -\frac{1}{4} \times \begin{cases} \sum_{j=1}^N d_{ij} & , t=1 \\ \sum_{j=1}^N d_{ij} + d_{ji} & , 2 \leq t \leq T-1 \\ \sum_{j=1}^N d_{ji} & , t=T \end{cases}$$

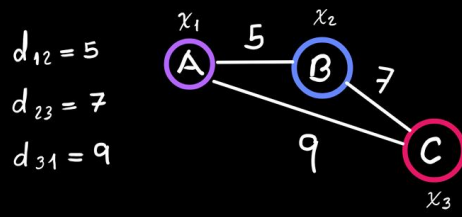
$$\mathcal{J}_{Kl} = \begin{pmatrix} 0 & \mathcal{J}_{12} & \mathcal{J}_{13} & \mathcal{J}_{14} & \mathcal{J}_{15} & \mathcal{J}_{16} & \mathcal{J}_{17} & \mathcal{J}_{18} & \mathcal{J}_{19} \\ \mathcal{J}_{21} & \uparrow & & & & & & & \\ \mathcal{J}_{31} & \text{restricci3n} & & & & & & & \\ \mathcal{J}_{41} & \text{especial} & & & & & & & \\ \mathcal{J}_{51} & & & & & & & & \\ \mathcal{J}_{61} & & & & & & & & \\ \mathcal{J}_{71} & & & & & & & & \\ \mathcal{J}_{81} & & & & & & & & \\ \mathcal{J}_{91} & & & & & & & & \end{pmatrix}$$

$i=1, t=1, t'=2$

$\mathcal{J}_{12} \Rightarrow K=1 = (i-1)T+t$   
 $l=2 = (j-1)T+t'+1$

$(x_{11} \cdot x_{12}) = 1$

|     |     |      |    |
|-----|-----|------|----|
|     | $t$ | $t'$ |    |
| $i$ | 1   | 1    | -1 |
|     | -1  | -1   | -1 |
|     | -1  | -1   | -1 |



con  $A = 100$   
 $B = 1$   
 $A \gg B_{m \times n}$

$$\mathcal{H} = \mathbf{Z}^T \mathbf{J} \mathbf{Z} + \mathbf{h}^T \mathbf{Z}$$

mismo  
 problema de  
 un punto de  
 vista diferente.

$$\mathbf{Z}_k = \begin{pmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \\ z_5 \\ z_6 \\ z_7 \\ z_8 \\ z_9 \end{pmatrix} \quad \mathbf{J}_{kl} = \begin{pmatrix} 0,000 & 50,000 & 50,000 & 50,000 & 1,250 & 0,000 & 50,000 & 2,250 & 0,000 \\ 50,000 & 0,000 & 50,000 & 1,250 & 50,000 & 1,250 & 2,250 & 50,000 & 2,250 \\ 50,000 & 50,000 & 0,000 & 0,000 & 1,250 & 50,000 & 0,000 & 2,250 & 50,000 \\ 50,000 & 1,250 & 0,000 & 0,000 & 50,000 & 50,000 & 50,000 & 1,750 & 0,000 \\ 1,250 & 50,000 & 1,250 & 50,000 & 0,000 & 50,000 & 1,750 & 50,000 & 1,750 \\ 0,000 & 1,250 & 50,000 & 50,000 & 50,000 & 0,000 & 0,000 & 1,750 & 50,000 \\ 50,000 & 2,250 & 0,000 & 50,000 & 1,750 & 0,000 & 0,000 & 50,000 & 50,000 \\ 2,250 & 50,000 & 2,250 & 1,750 & 50,000 & 1,750 & 50,000 & 0,000 & 50,000 \\ 0,000 & 2,250 & 50,000 & 0,000 & 1,750 & 50,000 & 50,000 & 50,000 & 0,000 \end{pmatrix} \quad \mathbf{h}_k = \begin{pmatrix} -103,500 \\ -107,000 \\ -103,500 \\ -103,000 \\ -106,000 \\ -103,000 \\ -104,000 \\ -108,000 \\ -104,000 \end{pmatrix}$$

¿Qué combinación, Qué serie, Qué secuencia,  
 me da la mínima energía?

si interpretamos  
 a  $\mathcal{H}$  como un  
 hamiltoniano  
 real

$$\mathbf{Z}_k \rightarrow \mathbf{Z}_{i,t}$$

Secuencia Espacial y temporal  
 para hallar  $E_{\min}$ .

pasemos al  
 lenguaje  
 cuántico



Hamiltoniano:

$$\mathcal{H} = \mathbf{z}^T \mathbf{J} \mathbf{z} + \mathbf{h}^T \mathbf{z}$$

$$\hat{H} = \mathcal{H} |z_1 z_2 \dots z_m\rangle$$

Podemos proyectar  
en  $\mathbf{z}$ .

Con  $z_k = \hat{z}_k$

$$\hat{z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Operador

$$\hat{H} = \sum_{1 \leq k < l \leq M} J_{kl} \hat{z}_k \otimes \hat{z}_l + \sum_{k=1}^M h_k \hat{z}_k$$

donde  $\mathbf{J}$  y  $\mathbf{h}$  ya los conocemos

↳ Como solo usamos

$$z_k \otimes z_k \otimes z_l$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

⇒  $\hat{H}$  es diagonal (en la base computacional  $\mathbf{v}_{\hat{z}} \otimes \mathbf{v}_{\hat{z}} \dots$ )

## Campos locales

$$h_{it} = \underbrace{B h_{it}^{(c)}}_{\text{término de costo}} + \underbrace{A(1-N/2)}_{\text{restricción temporal}} + \underbrace{A(1-\tau/2)}_{\text{restricción espacial}}$$

## Acoplamientos

• transición  $Q_c$ :

$$T_{(i,t)(j,t+1)} = \frac{B}{4} d_{ij}$$

para  $t=1, \dots, T-1, \forall i, j$

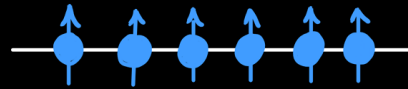
• mismo tiempo  $Q^{temp}$ :

$$T_{(i,t)(j,t)} = \frac{A}{2}$$

• mismo espacio  $Q^{esp}$ :

$$T_{(i,t)(i,t)} = \frac{A}{2}$$

un estado inicial

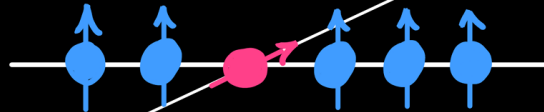


$\hat{\mathcal{H}}$  es un operador de evolución

$$\mathcal{H} \propto V_h(q_i) + V_J(q_i, q_j)$$

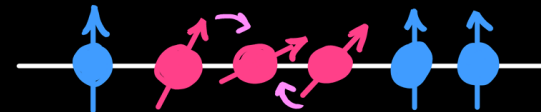
$$V_h(q_i) = h_i \cdot q_i$$

$$E_{ext} \propto h$$



tendencia individual a moverse

$$V_J(q_i, q_j) = J(q_i, q_j)$$

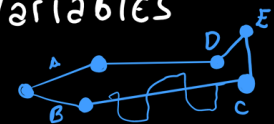


$$E_{int} \propto J$$

Coordinación/restricción entre variables

"acelerómetros"

combinación especial que te impulse



fren, pero con tiempo de espera

$$\hat{H} = \sum_{1 \leq k \leq l \leq M} J_{kl} \hat{Z}_k \otimes \hat{Z}_l + \sum_{k=1}^M h_k \hat{Z}_k$$

donde  $J_{kl}$  y  $h_k$  ya los conocemos  
Buscamos  $E_{\min}$ :

$$E_{\min}(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle$$

Pero,  $\nabla E_{\min} = 0$   $\overset{\text{sing}}{\text{?}}$

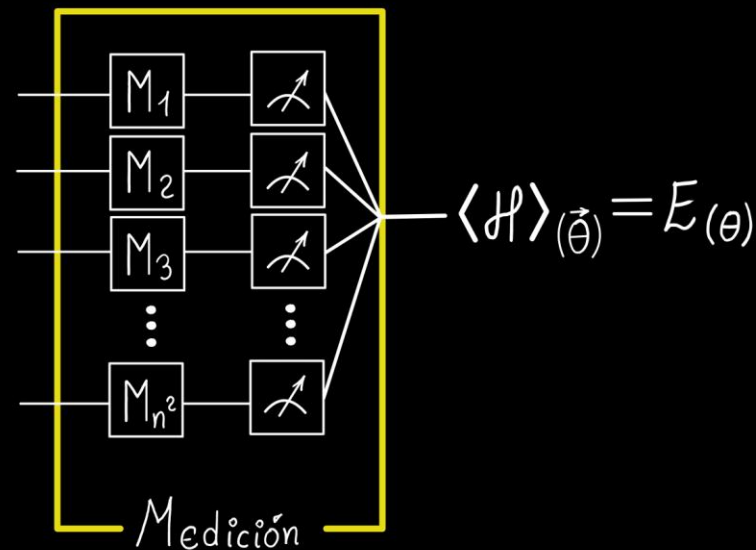
con:  $\hat{H} = \mathcal{H} |z_1 z_2 \dots z_m\rangle$   
podemos proyectar en  $\mathcal{Z}$ .

$|\psi\rangle = \sum_{\mathcal{Z}} c_{\mathcal{Z}} |\mathcal{Z}\rangle$ ,  $\sum_{\mathcal{Z}} |c_{\mathcal{Z}}|^2 = 1$   
 $2^M$  posibles configuraciones de  $\mathcal{Z}$   
 $|\mathcal{Z}\rangle = |z_1 z_2 z_3 \dots z_m\rangle$

$$\begin{aligned} \langle \psi | \hat{H} | \psi \rangle &= \sum_{\mathcal{Z}, \mathcal{Z}'} c_{\mathcal{Z}}^* c_{\mathcal{Z}'} \underbrace{\langle \mathcal{Z} | \hat{H} | \mathcal{Z}' \rangle}_{\mathcal{H}(\mathcal{Z}) \delta_{\mathcal{Z}\mathcal{Z}'}} \\ &\text{! } \hat{H} \text{ es diagonal! } \end{aligned}$$

Si  $\hat{H}$  no fuera diagonal  $\langle \hat{H} \rangle = \theta$

$\circ \mathcal{H}_{\min} = \langle \hat{H} \rangle_{\min}$   
 $\mathcal{H}_{\min} = E_{\min}$



Pero, ¿ $E_{\min}$  es solo el valor esperado?

Si pero del  $|\psi(\theta)\rangle$  correcto  
Por teorema,

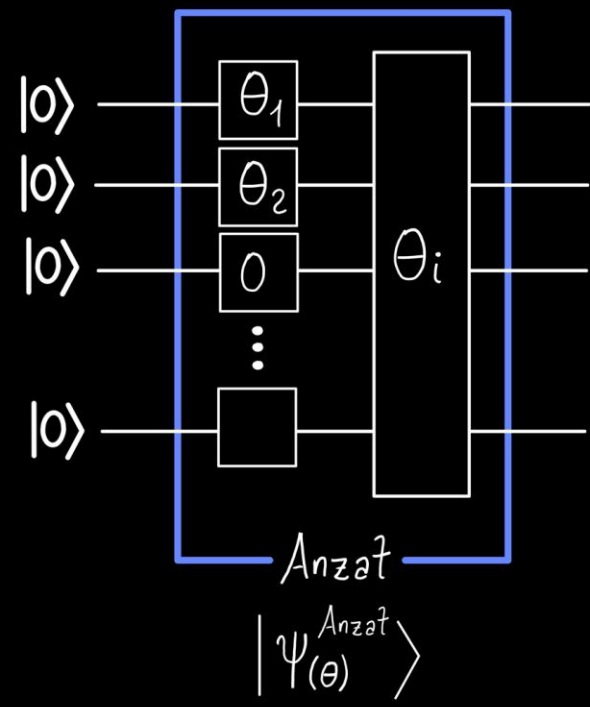
$$\langle \psi | \hat{H} | \psi \rangle \geq E_{\min}$$

La igualdad solo se cumple si  $|\psi\rangle$  es combinación lineal de estados base con  $E_{\min}$ .

VQE busca:

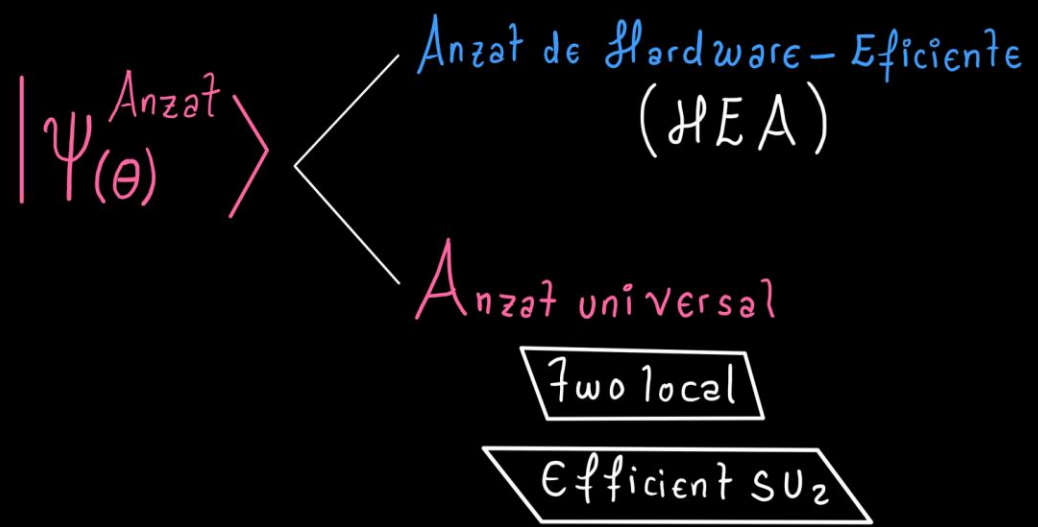
$$\min_{\theta} E(\theta) = \min_{\theta} \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle$$

$$E(\theta) = \sum_{1 \leq k < l \leq M} J_{kl} \langle \psi(\theta) | \hat{Z}_k \otimes \hat{Z}_l | \psi(\theta) \rangle + \sum_{k=1}^M h_k \langle \psi(\theta) | \hat{Z}_k | \psi(\theta) \rangle$$

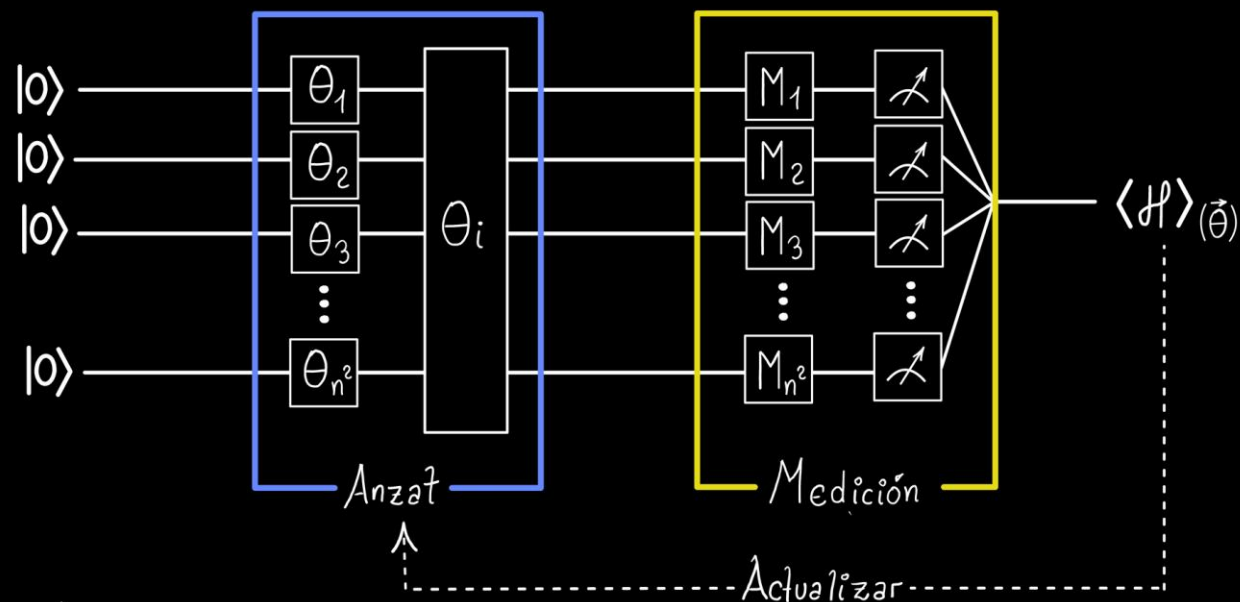


¿Pero qué  $|\psi(\theta)\rangle$  usamos?

esto es arbitrario, tenemos las sig propuestas:



Entonces ya tenemos:



Optimizador

$$\min_{\theta} f(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle$$

① Inicializar  $\theta$  aleatoriamente

② Iteraciones  $1, 2, \dots, \text{max\_iter}$

→ Evaluar  $f(\theta)$  **circuito cuántico + medición**

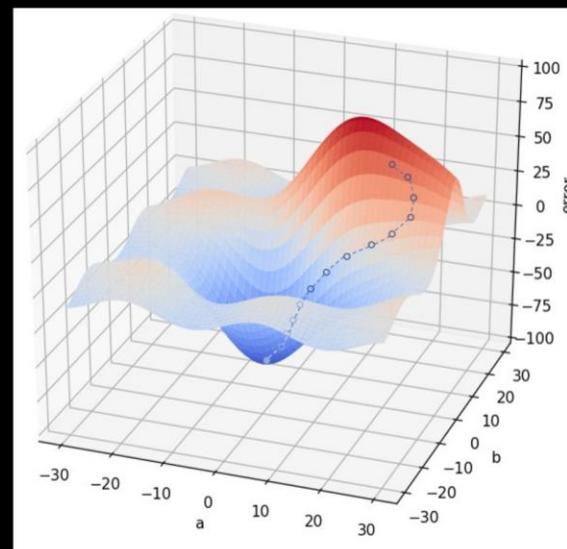
→ Actualizar según:

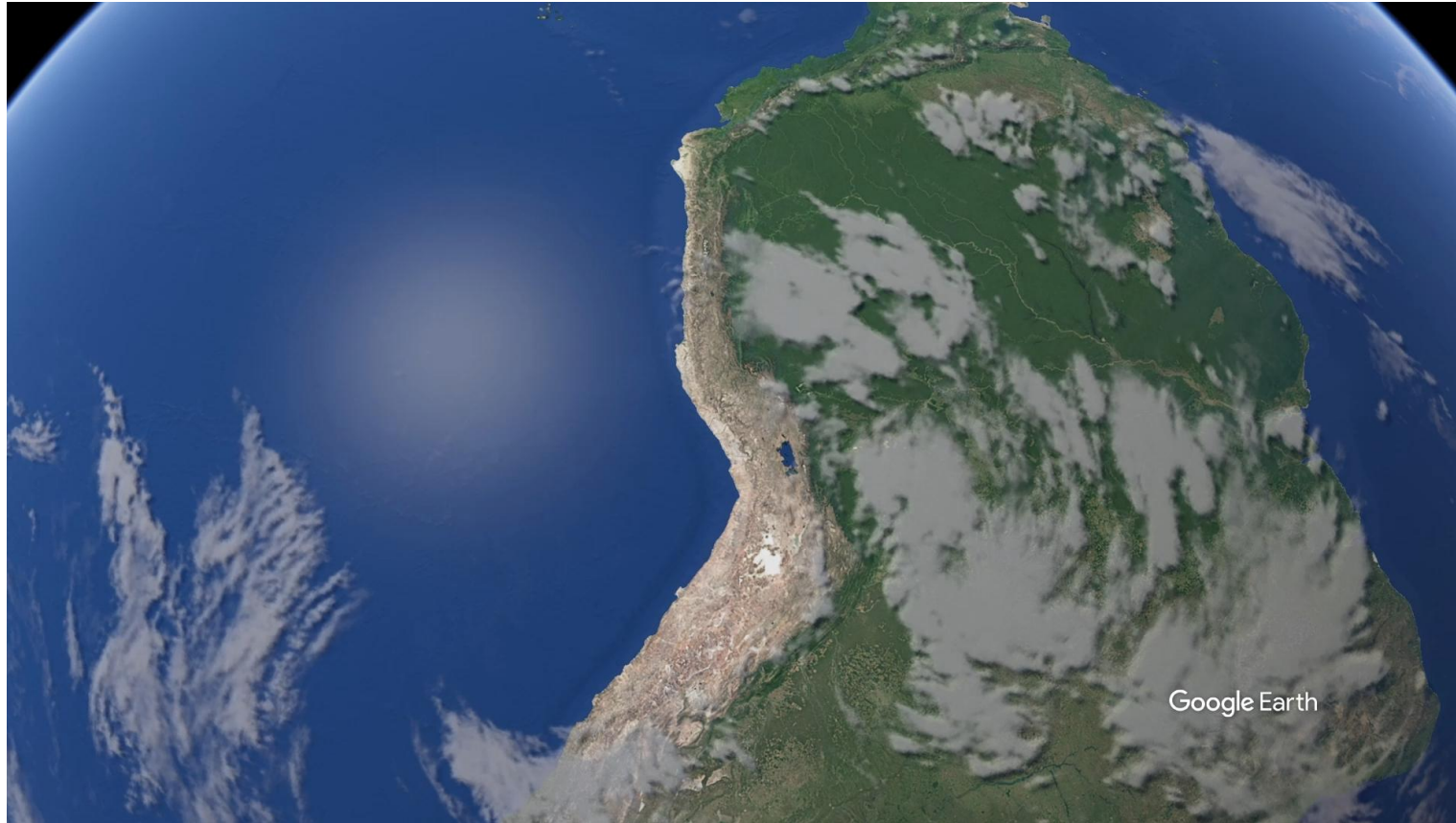
**Denso por gradiente**

$$\theta_{k+1} = \theta_k - \eta \nabla f(\theta_k)$$

↑  
paso o tasa de aprendizaje

→ Si hay convergencia: **parar**





# Orcopampa



Oro



Subterránea



3,800 msnm



Castilla, Arequipa



Inició en 1967



Busca en Google Earth



A

T

B



Busca en Google Earth



Google

100 %

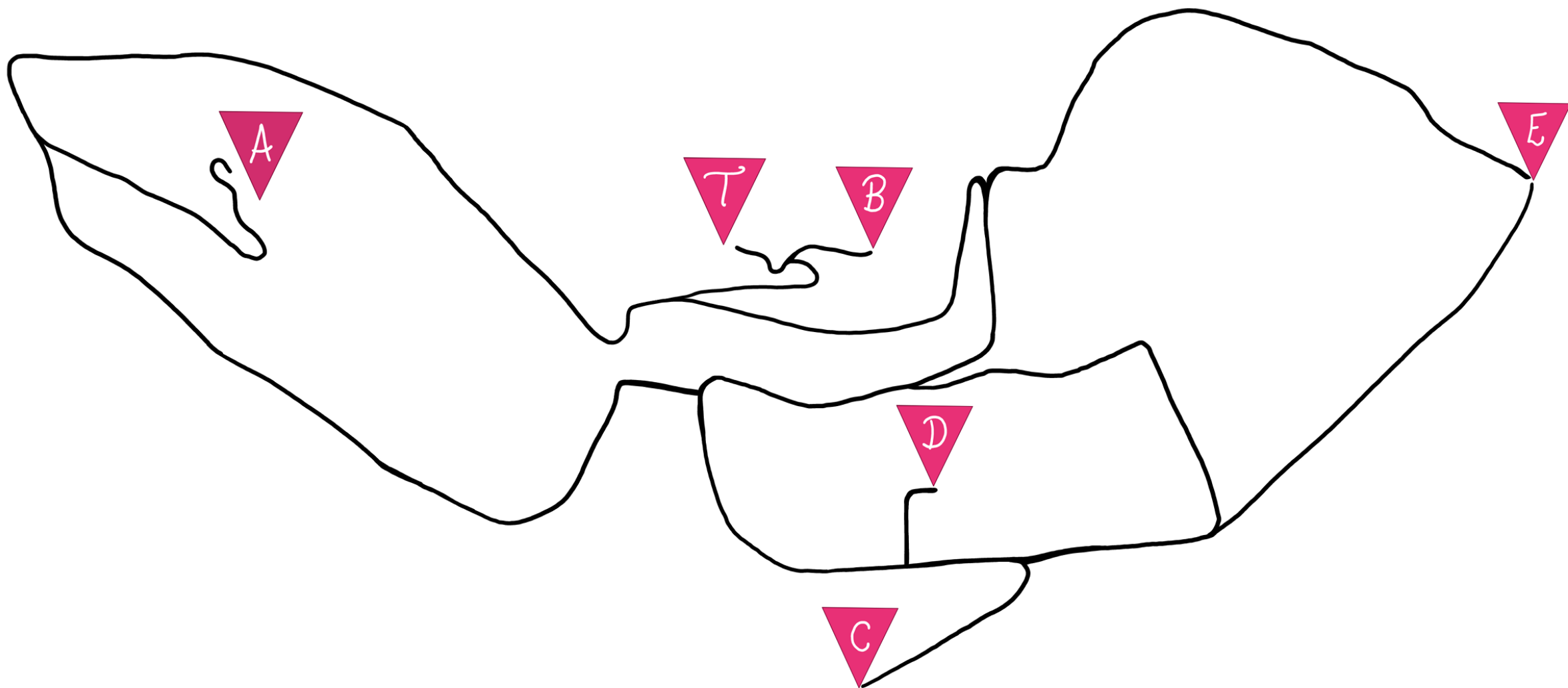
[Atribución de datos](#)

14/11/2023

200 m

Cámara: 5.520 m

15°16'33"S 72°19'58"W 3.795 m



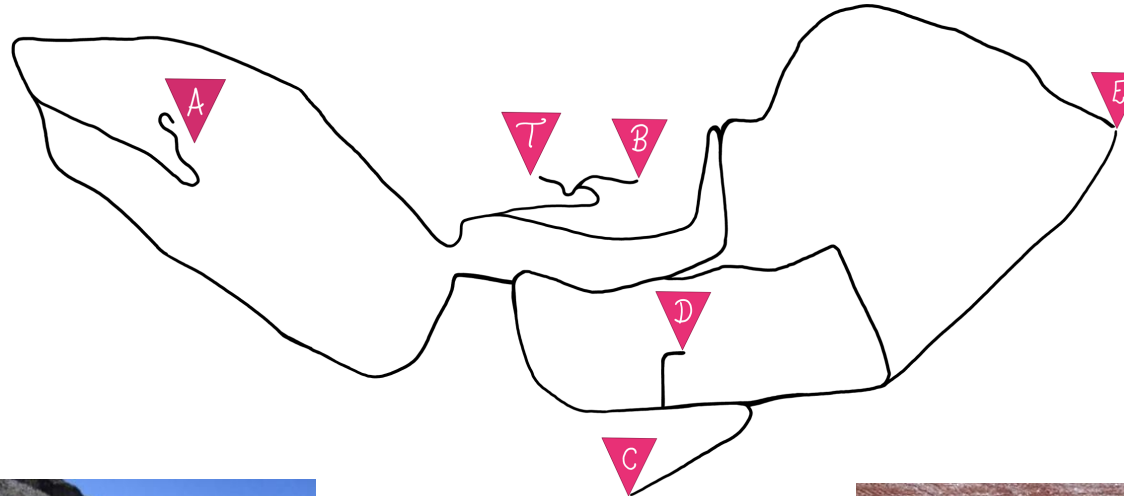


**Tajo (T):** Zona donde se hace la extracción. Se excava en bancos, hay rampas y de aquí salen los camiones cargados.



**Botadero (B):** Lugar donde se deposita material estéril que no sirve para procesar. Es “la papelera geológica” de la mina (pero con mucha ingeniería).

**Planta de procesamiento (A):** Es la instalación industrial donde el mineral extraído de la mina se transforma en un producto de valor comercial.



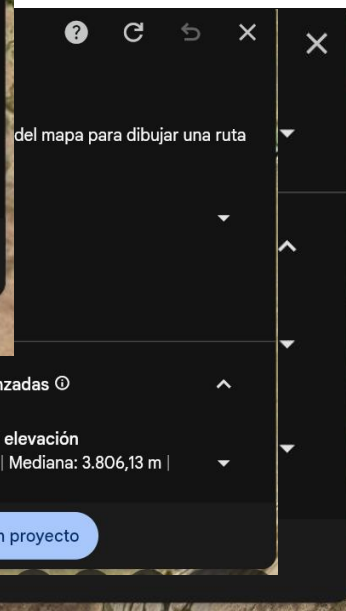
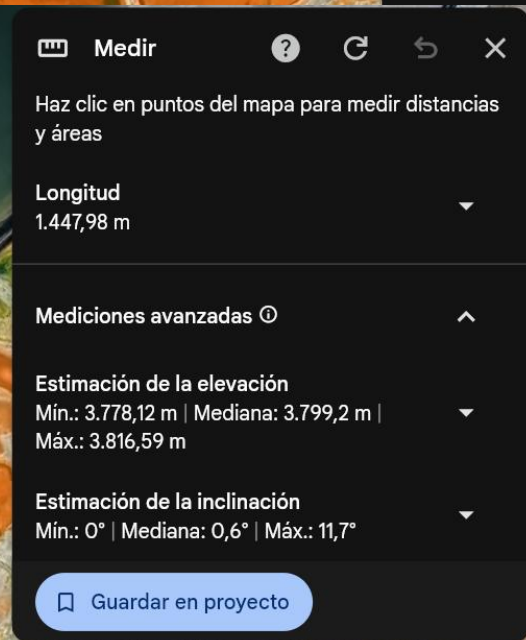
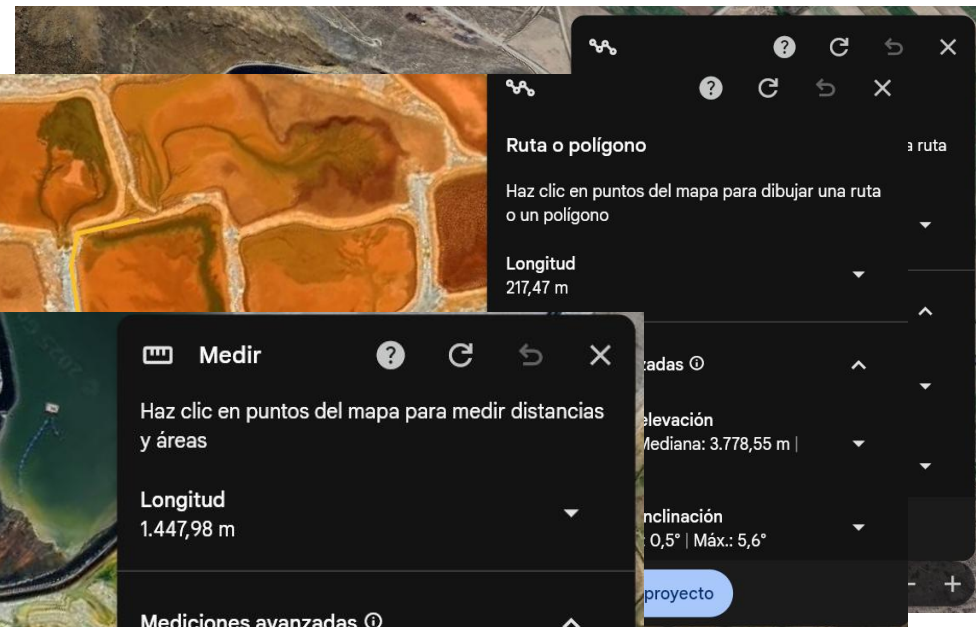
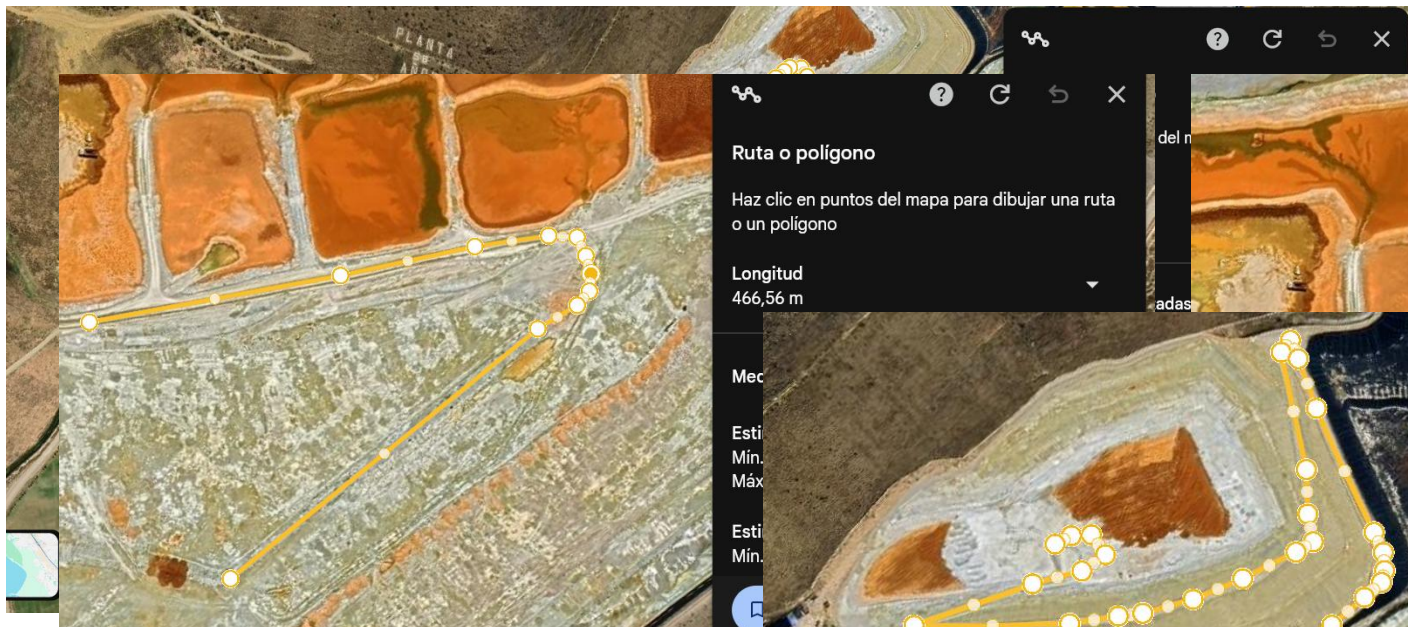
**Baja ley (C):** Mineral que sí contiene metal, pero tan poco que no conviene procesarlo hoy. Se guarda para cuando suba el precio del metal... o cuando el jefe esté optimista.

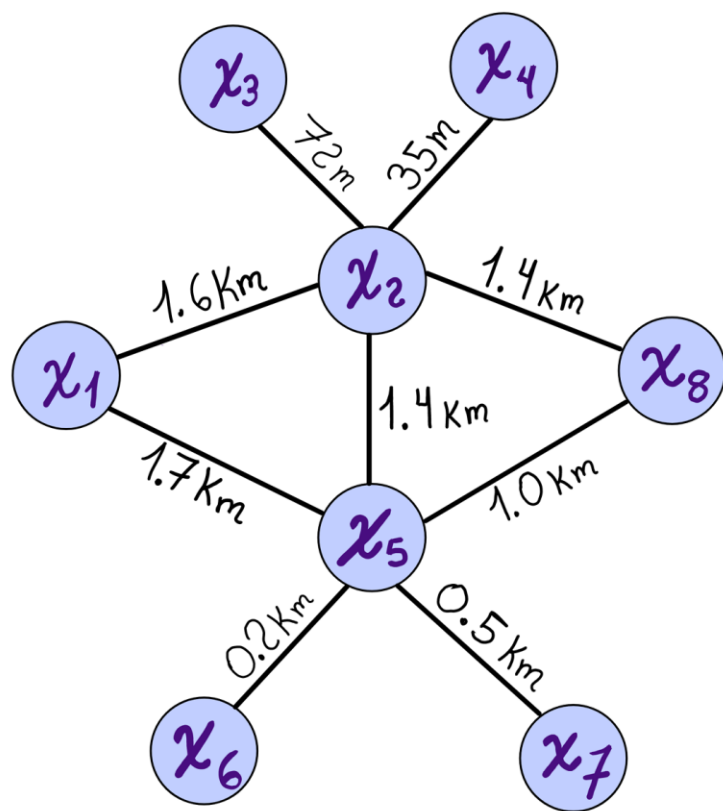
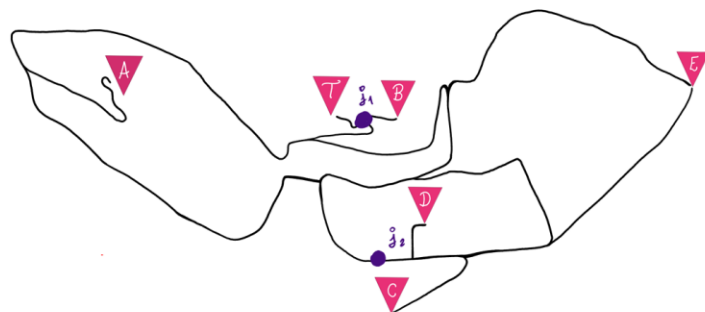


**Lixiviación (D):** Proceso donde un agente químico (por ejemplo cianuro o ácido) disuelve el metal.



**Relaves (E):** Restos sin valor después de procesar el mineral. Los relaves finos se envían a un embalse donde se decanta el agua.





$x_1$  : Planta de procesamiento

$x_2$  :  $j_1$

$x_3$  : Tajo

$x_4$  : Botadero

$x_5$  :  $j_2$

$x_6$  : Lixiviación

$x_7$  : Baja ley

$x_8$  : Relaves

$d_{12}$  (1,6 km)

$d_{23}$  (72 m)

$d_{24}$  (35 m)

$d_{28}$  (1,4 km)

$d_{25}$  (1,4 km)

$d_{15}$  (1,7 km)

$d_{85}$  (1,0 km)

$d_{56}$  (0,2 km)

$d_{57}$  (0,5 km)



GRACIAS